An area-law bound for SU(2) lattice gauge theory on a twisted small torus whose rate is the action of a flux-carrying column

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Abstract

We consider four-dimensional SU(2) Wilson lattice gauge theory on $T^2_N \times T^2_M$ with an 't Hooft twist $n_{12} = 1$ on the small torus T^2_N, and the bias $\theta(N, M, \beta) = (Z(0) - Z(1))/(Z(0) + Z(1))$ between the two sectors of the flux n_{34} through the large plane. For fixed N an area law $-\ln\theta \geq \frac{1}{2}e^{-2\beta N^2}M^2$ follows from a thin $\mathbb{Z}_2$ vortex sheet. We show, for every $N \geq 2$, block size $\ell \geq 2$, $\beta \geq 1$ and $M \in 2\ell\mathbb{N}$, that $\theta \leq \exp(-c\,\beta^{-a}e^{-\beta(S_*(N,\ell)+\varepsilon(\beta))}M^2)$ with $\varepsilon(\beta) = O(\sqrt{\ln\beta/\beta})$, where $S_*(N,\ell)$ is the minimal Wilson action of the n_{34}-twisted column $T^2_N \times [0,\ell]^2$ with its frame frozen to the twist-eater vacuum; c, a and the constant in ε depend on N and ℓ, and one constant (a rigidity constant of the frame) is shown to exist but is not explicit. The proof uses an exact column representation of θ, the chessboard estimate (quoted from the literature) applied only to the indicator of frames of large action, and a quadratic rigidity bound for frames near the two flat orbits that rests on the vanishing of a twisted adjoint cohomology group. At fixed M we also show the reverse bound, so that the exponential rate of $-\ln\theta$ in β lies in $[S_{\min}(N,M), S_*(N,\ell)]$ for $M \in 2\ell\mathbb{N}$. Explicit configurations give computed upper bounds $S_* \leq 0.6762\pi^2,\ 0.9193\pi^2,\ 0.9707\pi^2$ for $N = 2, 3, 4$, against $2N^2 = 8, 18, 32$. At the semiclassical level S_* is the action of a lattice fractional instanton confined to a column; without an identification $\lim_\ell S_*(N,\ell) = \lim_M S_{\min}(N,M)$, which we do not prove, the fractional-instanton action enters only as an upper bound on the rate. Nothing here is uniform in N, and nothing bears on the continuum limit or on a mass gap. Two finite algebraic steps are checked in Lean 4 with Mathlib; all analytic steps are on paper.

1 Setting

Motivation. Compactifying two directions of four-dimensional SU(2) Yang–Mills theory on a small torus with 't Hooft flux $n_{12} = 1$ removes the classical zero modes [tH81]. At the semiclassical level the cheapest objects carrying flux n_{34} through the remaining plane are fractional instantons of topological charge $\frac{1}{2}$ and action $4\pi^2/g^2 = \pi^2\beta$ ($\beta = 4/g^2$), localised in the plane [GAM, TU], and a dilute gas of them gives an area law with string tension proportional to $e^{-\pi^2\beta}$ up to a prefactor [TU]. This paper asks how much of this picture can be proved on the lattice at fixed N, uniformly in the size M of the plane. The setting, the uniqueness of the twist-eater vacuum and the positivity of $S_{\min}$ below are also described, in Japanese, in [KSR].

Lattice and sectors. $\Lambda = (\mathbb{Z}/N)^2 \times (\mathbb{Z}/M)^2$, SU(2) link variables U, normalised Haar measure dU, plaquette action $S_p^z(U) = 1 - \frac{1}{2} z_p \operatorname{tr} U_p \in [0,2]$ and weight $e^{-\beta S^z(U)}$ with $S^z = \sum_p S_p^z$. The background twist $n_{12} = 1$ sets $z_p = -1$ on one $(1,2)$-plaquette in each $(1,2)$-plane; sector $n_{34} = 1$ in addition sets $z_p = -1$ on a sheet Π_q of the N^2 $(3,4)$-plaquettes over a unit square q of the plane. With $Z(0), Z(1)$ the two partition functions and $\Delta F = \ln(Z(0)/Z(1))$,

$$\theta(N, M, \beta) = \frac{Z(0) - Z(1)}{Z(0) + Z(1)} = \tanh \frac{\Delta F}{2}.$$

Through the Tomboulis–Yaffe inequality this bias controls Wilson loops in the large plane; we do not use that inequality. The twist-eater vacuum U^0 has holonomies $i\sigma_1, i\sigma_2$ around the two cycles of T_N^2, all links in directions $3, 4$ equal to 1, and $S^z(U^0) = 0$ in sector $n_{34} = 0$. In sector $(n_{12}, n_{34}) = (1,1)$ no configuration has action 0 (an element of SU(2) commuting with $i\sigma_1$ and $i\sigma_2$ is ± 1), so the minimal action $S_{\min}(N, M)$ of that sector is positive and at most $2N^2$.

What we call the rate. For fixed N and M, the rate is the exponential rate in β of $-\ln\theta$, i.e. the limits of $-\beta^{-1}\ln(-\ln\theta)$ as $\beta \to \infty$. A bound $-\ln\theta \geq c\, e^{-\beta S} M^2$ uniform in M is an area law at fixed N with rate at most S.

Limits stated in advance. (i) The identification of S_* with a fractional instanton is made at the semiclassical level only; the theorems are about the lattice quantities S_* and $S_{\min}$. (ii) No statement is uniform in N. (iii) The rigidity constant $c_R(N, \ell)$ is shown to exist; its value is not explicit, and so neither is the value of β beyond which Theorem A improves the thin-sheet bound. (iv) The identification $\lim_\ell S_*(N, \ell) = \lim_M S_{\min}(N, M)$ (property (G2) of §5) is not proved; wherever the rate is described as the fractional-instanton action, this is an upper bound unless (G2) is assumed, and we say so in the statement. (v) The chessboard estimate is quoted, not re-derived.

Blocks, frames, columns. Fix $\ell \geq 2$ and $M = K\ell$ with K even. The plane is tiled by closed squares D_t, $t \in T = (\mathbb{Z}/K)^2$, so $|T| = M^2/\ell^2$. A link is a *skeleton link* if its projection to the plane lies on the grid $\{x_3 \in \ell\mathbb{Z}\} \cup \{x_4 \in \ell\mathbb{Z}\}$. The *interior links* of t are the non-skeleton links projecting into D_t; there are $n_B = 2N^2(\ell-1)(2\ell-1)$ of them. The *frame* Σ_t is the set of skeleton links projecting into ∂D_t; as a cell complex it is $T_N^2 \times C_{4\ell}$, and its 2-cells are the *frame plaquettes*. Every plaquette either has only skeleton links, or has an interior link of exactly one block t and all its other links in Σ_t (Lemma 2.1). Write P_t for the latter plaquettes, $S_t^\pm(\Sigma, U_{\text{int}}) = \sum_{p \in P_t} S_p^{z^\pm}$ where z^- carries an extra sheet Π_{q_t} over a unit square $q_t \subset D_t$, and

$$Z_t^\pm(\Sigma) = \int \prod_{\text{int}(t)} dU\, e^{-\beta S_t^\pm}, \qquad \theta_t(\Sigma) = \frac{Z_t^+ - Z_t^-}{Z_t^+ + Z_t^-}, \qquad m_t^\pm(\Sigma) = \min S_t^\pm(\Sigma, \cdot).$$

$A_t(\Sigma)$ denotes the action of the frame plaquettes of t, and A_{sk} the action of the plaquettes all of whose links are skeleton links.

Definition 1.1. $S_*(N, \ell) := m_t^-(\Sigma^0)$, where Σ^0 is the restriction of U^0 to the frame: the minimal action of the n_{34}-twisted column with its frame frozen to the vacuum.

Constants: $c_0 = 8/(3\pi^3)$, $L_\partial = N^2(12\ell - 8)$, $L(\beta) = 4N^2\ell^2(12\ln\beta + 34) + 8\ln 2$.

2 Theorem A: the rate at fixed N

Theorem A (confinement direction)**.** *Let $N \geq 2$, $\ell \geq 2$, $\beta \geq 1$ and $M \in 2\ell\mathbb{N}$, and let $c_R = c_R(N,\ell) > 0$ be the rigidity constant of Lemma 2.8 (shown to exist, not explicit). Then*

$$\theta(N,M,\beta) \leq \exp\Big(- \frac{M^2}{2\ell^2}\, c_0^{n_B} e^{-6n_B} \beta^{-3n_B} \exp\big(- \beta S_*(N,\ell) - L_\partial \sqrt{\beta L(\beta)/c_R}\big)\Big).$$

$S_*(N,\ell)$ may be replaced by the column action of any explicit configuration with vacuum frame.

Where $\theta > 0$ this reads $-\ln\theta \geq c(N,\ell)\beta^{-a} e^{-\beta(S_*(N,\ell)+\varepsilon(\beta))} M^2$ with $a = 3n_B$ and $\varepsilon(\beta) = L_\partial\sqrt{L(\beta)/(c_R\beta)}$.

2.1 Exact column representation

Lemma 2.1 (partition of plaquettes)**.** *Every plaquette lies either in P_{sk} (all links skeleton) or in exactly one P_t, and every $p \in P_t$ has all its links in* $\mathrm{int}(t) \cup \Sigma_t$.

Proof. The projection $\pi(p)$ to the plane is a vertex ($(1,2)$-plaquettes), an edge ($(\mu,3)$, $(\mu,4)$, $\mu \leq 2$) or a unit square ($(3,4)$), and the links of p project onto faces of $\pi(p)$. If $\pi(p)$ lies in the grid, all links are skeleton. Otherwise $\pi(p)$ lies in a unique closed square D_t, and p has a link whose projection leaves the grid (for a unit square, one of its four edges is off the grid because $\ell \geq 2$); the links of p projecting into the grid project into $D_t \cap \mathrm{grid} = \partial D_t$. □

Lemma 2.2 (central moves)**.** *(a) The partition function with sheets on Π_q, $q \in Q$, depends only on $|Q| \bmod 2$. (b) $Z_t^{\pm}$ and $m_t^{\pm}$ do not depend on the choice of $q_t \subset D_t$. (c) If $\Sigma^\flat$ is obtained from Σ by multiplying the 3-links over one edge of ∂D_t by -1, then $Z_t^s(\Sigma^\flat) = Z_t^{-s}(\Sigma)$ and $m_t^s(\Sigma^\flat) = m_t^{-s}(\Sigma)$.*

Proof. Multiplying the 3-links at $(x_1, x_2, a, b+1)$, for all (x_1,x_2), by the central element -1 preserves Haar measure and changes the sign of $\mathrm{tr}\, U_p$ exactly on $\Pi_{(a,b)} \cup \Pi_{(a,b+1)}$: a $(3,4)$-plaquette contains one such link if and only if it lies in these sheets, a $(\mu,3)$-plaquette with $\mu \leq 2$ contains two or none, and other plaquettes contain none. The same holds in direction 4, and the dual lattice is connected; this gives (a). In (b) the flipped links are interior links of t. In (c) the flipped links lie, among the plaquettes of P_t, only in the sheet over the unit square of D_t adjacent to the edge (the $(\mu,3)$-plaquettes over the edge are frame plaquettes and contain two flipped links); this adds one sheet, which (b) moves onto q_t, where two sheets cancel. The substitutions are bijections of the interior configurations, so the minima transform in the same way. □

Proposition 2.3 (column representation)**.** *Choose q_t reflection-covariantly (mirrored in adjacent blocks). Let $\tilde\mu(dU,s) \propto e^{-\beta S^{z^s}(U)} dU$ on configurations and sign vectors $s \in \{\pm\}^T$, where z^s carries a sheet over q_t for each t with $s_t = -$, and let ν be its marginal on skeleton links. Then*

$$\theta = \Big\langle \prod_{t\in T} s_t \Big\rangle_{\tilde\mu} = \mathbb{E}_\nu\Big[\prod_{t\in T} \theta_t(\Sigma)\Big].$$

Proof. By Lemma 2.2(a), $Z[s] = Z(0)$ if $\prod_t s_t = 1$ and $Z(1)$ otherwise, and half of the $2^{|T|}$ sign vectors have each parity; so $\sum_s (\prod_t s_t) Z[s] = 2^{|T|-1}(Z(0) - Z(1))$ and $\sum_s Z[s] = 2^{|T|-1}(Z(0)+Z(1))$. By Lemma 2.1, $Z[s] = \int d\Sigma\, e^{-\beta A_{\mathrm{sk}}(\Sigma)} \prod_t Z_t^{s_t}(\Sigma)$, and the sums over s factorise block by block: $\sum_s \prod_t s_t Z_t^{s_t} = \prod_t (Z_t^+ - Z_t^-)$ and $\sum_s \prod_t Z_t^{s_t} = \prod_t (Z_t^+ + Z_t^-)$ (Lean: `ColumnChessboard.signed_sum` and `unsigned_sum`). Divide. □

2.2 Reflection positivity and frames of large action

Proposition 2.4. *$\tilde\mu$ is reflection positive for the site reflections $x_3 \mapsto 2k\ell - x_3$ and $x_4 \mapsto 2k\ell - x_4$. Consequently, by the chessboard estimate [FILS, Sh, Bi], for bounded functions f_t of the variables of the closed block D_0, $|\langle \prod_t f_t \circ \vartheta_t\rangle_{\tilde\mu}| \le \prod_t \langle \prod_s f_t \circ \vartheta_s\rangle_{\tilde\mu}^{1/|T|}$, where ϑ_t is the composition of block reflections mapping D_t to D_0.*

Proof of reflection positivity. The reflection maps links of one closed half to the other (inverting links normal to the hyperplane), the sheet signs of mirrored blocks to each other, twist plaquettes to twist plaquettes, and sheets to sheets by the covariant choice of q_t; $\operatorname{tr} U_p^{-1} = \operatorname{tr} U_p$ in SU(2), and Haar measure is inversion invariant. Every plaquette lies in a fixed hyperplane or in one closed half (sheets are $(3,4)$-plaquettes, never inside a hyperplane x_3 or $x_4 =$ const). So the weight factorises as $A(\phi)F(\phi,\psi_+,s_+)F(\phi,\Theta\psi_-,\Theta s_-)$ with ϕ the variables in the fixed hyperplanes and $A, F \ge 0$, and $\langle G\,\Theta\bar G\rangle = \tilde Z^{-1}\int d\phi\, A(\phi)\,|\sum_{s_+}\int d\psi_+ FG|^2 \ge 0$. The chessboard estimate is the general consequence of reflection positivity for a family of reflections through all block-boundary hyperplanes [FILS, Sh, Bi]; it needs $K = M/\ell$ even. We quote it and do not re-derive it. □

Lemma 2.5 (reduction to frames of large action)**.** *Let G_0 be an event on the frame of block 0, $G_t = \vartheta_t^{-1}G_0$, and suppose $|\theta_t| \le 1-\eta$ on G_t with $\eta \in [0,1]$. With $p = \nu(\bigcap_t G_t^c)^{1/|T|}$, $\theta \le (1-\eta(1-p))^{|T|}$.*

Proof. $|\theta_t| \le (1-\eta) + \eta\mathbf{1}_{G_t^c}$ pointwise. Expand the product over subsets $S \subset T$ (Lean: `ColumnChessboard.prod_expand`); by Proposition 2.4 applied to $f_t = \mathbf{1}_{G_0^c}$ for $t \in S$ and $f_t = 1$ otherwise, $\nu(\bigcap_{t\in S} G_t^c) \le p^{|S|}$; sum the binomial series (Lean: `ColumnChessboard.binomial_bound`). □

Lemma 2.6 (frames of large action are rare)**.** *Let $\varepsilon_1 = L(\beta)/\beta$ and $G_t = \{A_t \le \varepsilon_1\}$. Then $p \le \frac12$ for all $\beta \ge 1$.*

Proof. A skeleton plaquette belongs to at most four frames (the $(1,2)$-plaquettes over a grid corner), so on $\bigcap_t G_t^c$ we have $A_{\mathrm{sk}} \ge \frac14\sum_t A_t > \varepsilon_1|T|/4$. By Markov's inequality $\nu(\bigcap_t G_t^c) \le e^{-\beta\varepsilon_1|T|/4}\mathbb{E}_\nu e^{\beta A_{\mathrm{sk}}}$, and $\mathbb{E}_\nu e^{\beta A_{\mathrm{sk}}} = \Xi^{-1}\int d\Sigma \prod_t (Z_t^+ + Z_t^-) \le 2^{|T|}/\Xi \le 2/Z(0)$, because $Z_t^\pm \le 1$ and $\Xi = 2^{|T|-1}(Z(0)+Z(1))$. Each S_p is 1-Lipschitz in each of its links in operator norm ($\frac12|\operatorname{tr}(A-B)| \le \|A-B\|$ and $\|\prod U_i - \prod V_i\| \le \sum\|U_i - V_i\|$ for unitaries). On the set where every link is within $r = 1/\beta$ of U^0 the action is at most $4\cdot 6N^2M^2\,r$. The class angle $\psi \in [0,\pi]$ of $W^{-1}U$ has density $\frac{2}{\pi}\sin^2\psi$, and $\|U-W\| = 2\sin(\psi/2) \le \psi$; so a ball of radius $r \le \pi/2$ has Haar measure at least $\frac{2}{\pi}\int_0^r (2\psi/\pi)^2 d\psi = c_0 r^3$. Hence $Z(0) \ge (c_0\beta^{-3})^{4N^2M^2}e^{-24N^2M^2} \ge \exp(-N^2M^2(12\ln\beta + 34))$, using $4\ln(1/c_0) < 10$. With $M^2 = \ell^2|T|$, $p \le 2^{1/|T|}\exp(-L(\beta)/4 + N^2\ell^2(12\ln\beta+34)) = 2^{1/|T|}\cdot\frac14 \le \frac12$. □

2.3 Flat frames and rigidity

Lemma 2.7 (flat frames)**.** *$A_t(\Sigma) = 0$ if and only if Σ lies in one of two distinct gauge orbits, that of Σ^0 or that of $\Sigma^\flat$ (Σ^0 modified as in Lemma 2.2(c)). They are distinguished by the sign ±1 of the holonomy around ∂D_t.*

Proof. On each slice $T_N^2 \times \{c\}$ a flat twisted configuration is, in a comb gauge, determined by two holonomies with $P_1P_2P_1^{-1}P_2^{-1} = -1$. For unit quaternions, $P_1P_2 = -P_2P_1$ forces both to be purely imaginary and orthogonal, so (P_1,P_2) is conjugate to $(i\sigma_1, i\sigma_2)$ with stabiliser $\{\pm1\}$. Gauge every slice to the twist eater. Flatness of the mixed plaquettes over a cycle edge τ gives $V(x+\hat\mu) = (U^0_{x,\mu})^{-1}V(x)U^0_{x,\mu}$ for the links $V(x)$ along τ; going around the two cycles of T_N^2 shows that $V(0)$ commutes with $i\sigma_1$ and $i\sigma_2$, hence $V \equiv \epsilon_\tau \in \{\pm1\}$. A gauge

transformation -1 on one slice flips ϵ on its two cycle edges, so only $\prod_\tau \epsilon_\tau$, which is $\frac{1}{2}\operatorname{tr}$ of the holonomy around ∂D_t, is invariant. □

Lemma 2.8 (rigidity)**.** *There is $c_R = c_R(N,\ell) > 0$ such that $A_t(\Sigma) \geq c_R\, d(\Sigma)^2$ for every frame Σ, where d is the max-link operator-norm distance to the union of the two flat orbits.*

Proof. Write $\Sigma_e = e^{a_e}\Sigma^f_e$ near a flat frame Σ^f, $a \in \mathfrak{su}(2)^{\Sigma_t}$. Then $z_p U_p = \exp((d^1_f a)_p + O(|a|^2))$ with d^1_f the coboundary of the cellular complex of $T^2_N \times C_{4\ell}$ with coefficients in the adjoint local system of Σ^f, and $1 - \frac{1}{2}\operatorname{tr} e^X = \frac{1}{2}|X|^2 + O(|X|^4)$; so $A_t = \frac{1}{2}|d^1_f a|^2 + O(|a|^3)$. The tangent space of the orbit is $\operatorname{im} d^0_f$, so the Hessian is non-degenerate on the normal space if and only if $H^1 = 0$. In the gauge of Lemma 2.7 the cycle links are ± 1, which act trivially in the adjoint, so the local system is pulled back from $V = \mathrm{ad}(\text{twist eater})$ on T^2_N, and the cellular complex is the tensor product of the twisted complex of T^2_N and the untwisted complex of $C_{4\ell}$ (Künneth over $\mathbb{R}$). $H^0(T^2_N; V) = 0$: no vector of $\mathfrak{su}(2)$ is fixed by both $\mathrm{Ad}(i\sigma_1)$ and $\mathrm{Ad}(i\sigma_2)$. $H^2(T^2_N; V) \cong H_0(T^2_N; V)$ is the space of coinvariants, which is 0 because the images of $\mathrm{Ad}(i\sigma_1) - 1$ and $\mathrm{Ad}(i\sigma_2) - 1$ span $\mathfrak{su}(2)$. The Euler characteristic gives $\dim H^0 - \dim H^1 + \dim H^2 = 3\chi(T^2) = 0$, so $H^1(T^2_N; V) = 0$, and by Künneth $H^1(T^2_N \times C_{4\ell}; V) = 0$. Near the compact zero set the Taylor expansion with remainder uniform on the zero set gives $A_t \geq \frac{1}{4}\lambda \operatorname{dist}^2$ with $\lambda > 0$ the minimum over the zero set of the smallest normal eigenvalue (this is the Morse–Bott situation [Bott]); away from a neighbourhood A_t is bounded below by compactness, while $d \leq 2$. Riemannian and max-link distances are comparable on the compact product. □

Remark 2.9. The radius of the neighbourhood and the bound away from it are not explicit; only their existence is used.

2.4 Proof of Theorem A

Lemma 2.10 (ball bound)**.** *For every frame and $\beta \geq 1$, $1 - |\theta_t| \geq \min(Z^+_t, Z^-_t) \geq c_0^{n_B}\beta^{-3n_B}e^{-6n_B}e^{-\beta \max_s m^s_t(\Sigma)}$.*

Proof. $1 - |\theta_t| = 2\min(Z^\pm_t)/(Z^+_t + Z^-_t)$ and $Z^\pm_t \leq 1$. Around a minimiser, on the set where each interior link is within $1/\beta$ of it, $S^s_t \leq m^s_t + 6n_B/\beta$ (each interior link lies in six plaquettes, all in P_t), and this set has Haar measure at least $(c_0\beta^{-3})^{n_B}$. □

Lemma 2.11 (frames of small action)**.** *If $A_t(\Sigma) \leq \varepsilon$ then $\max_s m^s_t(\Sigma) \leq S_*(N,\ell) + L_\partial\sqrt{\varepsilon/c_R}$.*

Proof. m^s_t is invariant under gauge transformations of the frame (extend by 1 to interior sites; by Lemma 2.1 every plaquette of P_t has its links in $\operatorname{int}(t) \cup \Sigma_t$). By Lemma 2.8 the frame is, after a gauge transformation, within $d = \sqrt{\varepsilon/c_R}$ of Σ^0 or $\Sigma^\flat$ in every link. m^s_t is L_∂-Lipschitz in the frame: L_∂ counts the incidences between plaquettes of P_t and frame links, namely $8N^2(\ell-1)$ mixed plaquettes (μ, τ') with τ' an inward edge at a non-corner boundary vertex, each with one frame link, and $4\ell N^2$ incidences of $(3,4)$-plaquettes with boundary edges. Finally $m^+_t(\Sigma^0) = 0$, $m^-_t(\Sigma^0) = S_*$, and $m^\pm_t(\Sigma^\flat) = m^\mp_t(\Sigma^0)$ by Lemma 2.2(c). □

Proof of Theorem A. Take $G_t = \{A_t \leq \varepsilon_1\}$, which is reflection covariant because the frame action is invariant under block reflections. By Lemmas 2.10 and 2.11, $|\theta_t| \leq 1 - \eta$ on G_t with

$$\eta = c_0^{n_B}\beta^{-3n_B}e^{-6n_B}\exp\big(-\beta S_* - L_\partial\sqrt{\beta L(\beta)/c_R}\big) \in [0,1].$$

By Lemma 2.6, $p \leq \frac{1}{2}$, and Lemma 2.5 gives $\theta \leq (1-\eta/2)^{M^2/\ell^2} \leq \exp(-\eta M^2/(2\ell^2))$ (Lean: `ColumnChessboard.exp_bound`). If an explicit configuration with vacuum frame and column action $\bar S$ replaces the minimiser in Lemma 2.11, the same proof gives the bound with $\bar S$. □

3 Theorem B and Corollary C: the reverse bound at fixed M

Theorem B (fixed-M reverse bound)**.** *For all $N, M \geq 2$ and $\beta \geq 1$ let $x = 2\exp\big(-\beta S_{\min}(N,M) + N^2M^2(12\ln\beta + 34)\big)$. If $x \leq \frac{1}{2}$ then $0 < \theta$ and $-\ln\theta(N,M,\beta) \leq 2x$.*

Proof. $\theta = 1 - 2Z(1)/(Z(0)+Z(1)) \geq 1 - 2Z(1)/Z(0)$. Every configuration of sector $(1,1)$ has action at least $S_{\min}(N,M)$, so $Z(1) \leq e^{-\beta S_{\min}(N,M)}$ (normalised Haar measure), and $Z(0) \geq \exp(-N^2M^2(12\ln\beta+34))$ as in the proof of Lemma 2.6. So $\theta \geq 1 - x \geq \frac{1}{2}$ and $-\ln\theta \leq -\ln(1-x) \leq x/(1-x) \leq 2x$. □

Corollary C. *For fixed $N \geq 2$, $\ell \geq 2$ and $M \in 2\ell\mathbb{N}$,*

$$S_{\min}(N,M) \leq \liminf_{\beta\to\infty} \frac{-\ln(-\ln\theta)}{\beta} \leq \limsup_{\beta\to\infty} \frac{-\ln(-\ln\theta)}{\beta} \leq S_*(N,\ell).$$

The best choice is $\ell = M/2$. The two ends are lattice quantities; reading the right end as a fractional-instanton action is a statement at the semiclassical level.

Proof. For large β, $x \leq \frac{1}{2}$ and Theorem B gives

$$-\beta^{-1}\ln(-\ln\theta) \geq S_{\min}(N,M) - O_{N,M}(\ln\beta/\beta).$$

Theorem A gives $-\beta^{-1}\ln(-\ln\theta) \leq S_*(N,\ell) + \varepsilon(\beta) + O_{N,\ell}(\ln\beta/\beta)$ with $\varepsilon(\beta) \to 0$. □

The ends are consistent: $S_{\min}(N,k\ell) \leq S_*(N,\ell)$ for all $k \geq 1$, because extending a column configuration with vacuum frame by the vacuum gives a configuration of sector $(1,1)$ on $T^2_N \times T^2_{k\ell}$ with the same action (Lemma 2.1; for $k = 1$ the opposite frame sides are identified consistently because the vacuum frame is translation invariant). The fixed-M bound loses $e^{O(M^2\ln\beta)}$ through the lower bound on $Z(0)$, so it says nothing uniformly in M; the M-uniform reverse bound is open (§5).

4 The thin-sheet bound and computed upper bounds for S_*

Proposition 4.1 (thin-sheet bound)**.** $|\theta_t| \leq \tanh(\beta N^2)$ *for every frame. Hence for $M \in 2\mathbb{N}$, with $\ell = 2$,*

$$-\ln\theta \geq \frac{M^2}{4}\big(-\ln\tanh(\beta N^2)\big) \geq \frac{M^2}{2}e^{-2\beta N^2}.$$

Proof. $|S_t^- - S_t^+| = |\sum_{p\in\Pi_{q_t}} \operatorname{tr} U_p| \leq 2N^2$, so $e^{-2\beta N^2} \leq Z_t^-/Z_t^+ \leq e^{2\beta N^2}$ and $|\theta_t| \leq \tanh(\beta N^2)$. Proposition 2.3 (no chessboard estimate is needed) gives $\theta \leq \tanh(\beta N^2)^{M^2/4}$, and $-\ln\tanh y = \ln\frac{1+u}{1-u} \geq 2u$ with $u = e^{-2y}$. □

Theorem A improves the exponent by $2N^2 - S_*$ per unit of β, but its prefactor is $\beta^{-3n_B}e^{-6n_B}e^{-O(\sqrt{\beta\ln\beta})}$ with a non-explicit constant, so it beats Proposition 4.1 only for β beyond a finite, large and non-explicit value.

Elementary properties of S_*. $S_*(N,\ell)$ is attained (compactness), is at most $2N^2$ (vacuum interior and one thin sheet), is non-increasing in ℓ (place a minimiser in a corner of a larger column and the vacuum elsewhere; every additional plaquette has all its links at vacuum values), and satisfies $S_*(N,\ell) \geq S_{\min}(N,k\ell)$ for $k \geq 1$. Hence $\lim_\ell S_*(N,\ell)$ exists and is at least $\limsup_k S_{\min}(N,k\ell)$ for each ℓ.

Computed upper bounds. Column minimisers were searched by cooling from a thin sheet plus noise; saved configurations were normalised exactly, the frame was checked to equal the vacuum exactly, and the column action was re-evaluated in 40-digit arithmetic. Each value is the action of an explicit configuration and therefore an upper bound on S_*; by the last sentence of Theorem A it can be inserted there. The values below are computed and were not recomputed for this paper.

(N,ℓ)	computed upper bound on $S_*(N,\ell)$	$/\pi^2$
$(2,12)$	6.67348665542	≤ 0.6762
$(3,12)$	9.07284242009	≤ 0.9193
$(4,8)$	9.57961427901 (search not converged)	≤ 0.9707

Against the thin-sheet rate $2N^2 = 8, 18, 32$, these give exponent gains of about 1.33, 8.93 and at least 22.4. Cooling on the periodic torus gives

$$S_{\min}(2,M) \approx 6.740117,\ 6.677345,\ 6.673724,\ 6.673486,\ 6.673485 \quad (M = 4,6,8,12,16)$$

and $S_{\min}(3,16) = 9.072494$; these cooled values are upper bounds on $S_{\min}$, not lower bounds, so they do not supply the left end of Corollary C as a number. With the computed upper bound $S_*(2,8) \le 6.67400717258$, Corollary C for $N = 2$, $M = 16$ places the rate in $[S_{\min}(2,16),\ 6.67400717259]$, where the cooled value of $S_{\min}(2,16)$ is 6.673485. That $S_*(N,\ell)$ approaches the periodic-torus values as ℓ grows, with an excess decaying roughly like $e^{-m_N \ell}/\ell$, $m_N = 2\,\mathrm{arsinh}\sin(\pi/2N)$, is a numerical observation.

5 What is not claimed

1. *(G1) Reverse bound uniform in M.* A dilute-gas lower bound

   $$\theta \ge \exp\big(-C\beta^{a'} e^{-\beta(S_* - o(1))} M^2\big)$$

   is open. One route is to rewrite θ exactly over connected clusters of frames of large action and to show that each cluster factor is positive with $1 - 2q_R \ge e^{-C|R|}$ (a property we call (Pos)); neither the rewriting nor (Pos) is claimed here.

2. *(G2) Identification of the rate.* $\lim_\ell S_*(N,\ell) = \lim_M S_{\min}(N,M)$ is not proved; only the inequality $\ge \limsup_k S_{\min}(N,k\ell)$ is. Without (G2), the statement "the rate is set by fractional instantons" holds only on the upper side: the rate is at most the confined-column action S_*.

3. *(G3) Large N.* $S_{\min}(N,\infty)/\pi^2 \approx 0.676,\ 0.919,\ 0.96$ for $N = 2,3,4$, approaching the continuum value 1, is numerical only. Nothing is uniform in N, and the passage from N small compared with the correlation length to N large is not addressed.

4. *(G4) Constants.* c_R and hence the range of β in which Theorem A improves Proposition 4.1 are not explicit; the prefactor β^{-3n_B} and the $\sqrt{\beta \ln \beta}$ term come from crude ball estimates and worst-case frames.

5. *(G5) From an area law to a gap.* An area law for the twist-summed bias does not by itself give a mass gap: on a product of a confining theory with a massless free field, Wilson loops of the first factor obey an area law while the spectrum has no gap. Any passage from area law to gap has to restrict the observables, and we do not attempt it.

Moreover: no continuum limit is taken; the semiclassical identification of S_* with a fractional instanton is interpretation, not a theorem here; the chessboard estimate is quoted; the computed values are not machine-checked.

6 What is machine-checked

One Lean 4 module with Mathlib, `ColumnChessboard`, checks the finite algebra used above and nothing analytic (declarations below are in that namespace):

declaration	role in this paper
`signed_sum`	$\sum_s \prod_t s_t Z_t^{s_t} = \prod_t (Z_t^+ - Z_t^-)$ (Prop. 2.3)
`unsigned_sum`	$\sum_s \prod_t Z_t^{s_t} = \prod_t (Z_t^+ + Z_t^-)$ (Prop. 2.3)
`prod_expand`	expansion of $\prod_t ((1-\eta) + \eta x_t)$ over subsets (Lemma 2.5)
`binomial_bound`	from $a_S \le p^{\lvert S\rvert}$ to $(1-\eta+\eta p)^{\lvert T\rvert}$, for $0 \le \eta \le 1$ (Lemma 2.5)
`exp_bound`	$(1-y)^n \le e^{-ny}$ for $y \le 1$ (proof of Theorem A)

For each of the five, `#print axioms` reports `[propext, Classical.choice, Quot.sound]` only; `sorry` and `native_decide` are not used. Measures, the chessboard estimate, Haar-ball volumes, the cohomology computation, the Morse–Bott argument and all numerical values are outside Lean. The module is distributed at `https://computoergosum.com/principia/lean/principia-ym-fractional-instanton-rate-2026-10-15.tar.gz` (with a `.sha256` file at the same address with that suffix), and the `#print axioms` output at `https://computoergosum.com/principia/lean/axioms-ym-fractional-instanton-rate-2026-10-15.txt`; the ledger and build instructions of the site's Lean developments are at `https://computoergosum.com/en/principia/lean/index.html`.

To verify: unpack, run `lake exe cache get`, then `LEAN_NUM_THREADS=1 lake env lean ColumnChessboard.lean`; the `#print axioms` lines at the end of the module should agree with the report above and with the distributed axiom log.

7 Disclosure of AI use

This paper was produced with the assistance of AI agents. The constructions, the informal proofs, the computations, the Lean 4 development including the formalised statements, the literature search, the reports on which this paper is based, and the text itself were written by instances of Claude (Anthropic) run as the coding agent Claude Code, across many separate sessions, each of which began without memory of the previous ones and read the written record instead; the proofs of Sections 2–4 were re-derived line by line from an earlier written record before this text was written. A second group of instances, running on the server of the website named on the title page, is responsible for rebuilding the Lean development from the distributed archive on a separate machine, for checking the claims of this paper against the reports, and for preparing the distributed version; the independent rebuild was completed on 2026-10-15 (every file checked with `lake env lean` in the same environment on a separate session, rc=0, standard axioms only). The human author set the task, fixed the working rules and load limits, reviewed the record, and decides whether and where the result is reported; he is responsible for this paper. Adversarial review was carried out only by further instances of the same model family, which share the same blind spots. This paper should be read as a verified candidate, not as a refereed result: what the Lean kernel checks, it checks; everything else rests on the author's word. In particular the analytic steps, the quoted chessboard estimate and all numerical values are not machine-checked. This edition and the Lean development it cites are kept in the repository `https://github.com/yoshikazuhosomura-sys/principia` (the paper under `papers/ym-fractional-instanton-rate`, the development under `lean/ym-fractional-instanton-rate`). The web edition of this paper, published on the site named on the title page, lists the same work under the names Kiichi, Shiori and Rin, the human author and the two groups of AI agents; the two editions differ only in the title page and in this section.

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